

**Dota 2' guessing reward**

New Dota Pro Circuit app lets you win Shards by predicting the winners of Circuit matches <https://dgit.in/apr19-37>

**Apple Watch irregular heart beats**

Stanford's Apple Watch-based heart study appears to have been a success, with a few caveats. <https://dgit.in/apr19-38>

Engage all Defences

India's defence startups are aiding our armed forces and strengthening our borders

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The strength of a country's defences is measured in many ways. While numbers definitely play a big role, current technical and scientific advancements that aid the defence forces are the deciding factor more often than not. To stay ahead of the game in this aspect, a country needs a good research, development and manufacturing ecosystem. In recent times, a major part of this ecosystem has been startups. Quite interestingly, the Government of India has also been encouraging startups to take up the mantle of innovation when it comes to our defence capabilities.

EFFORTS FROM THE GOVERNMENT

Last June, the defence ministry had set up a new set of rules that allow homegrown startups to participate in military projects to streamline their attention on state of the art research and development. Under the new rules, startups recognised by the Department of Industrial Policy & Promotion (DIPP) under certain categories will automatically qualify to take part in specified defence projects. These

categories range from aeronautics, nanotechnology virtual reality, renewable technologies, robotics, green technologies and internet of things. With these rules specified, the three defence services could shortlist projects that can be awarded under the category. Among the 53 priority projects identified by The Army, Air Force and Navy the notable ones include Maneuverable Expendable Aerial Targets (MEAT) for Army Air Defence, lightweight body armour, a robotic surveillance platform, diesel engines for boats, limpet mines, air to ground rockets and long-range glide bombs.

On August 4, 2018, the Government of India also launched the Defence India Startup Challenge as a joint initiative of the Atal Innovation Mission, the Department of Industrial Policy and Promotion (DIPP), and the Defence Innovation Organisation (a ministry of defence initiative). From see-through armour, tactical 4G LAN to unmanned vehicles and laser weaponry, the program puts forth eleven challenges for Indian startups to solve. There are multiple other instances of the Government working towards bringing defence startups and the defence forces closer together. Some of the best

defence startups in India and their work is proof of that.

Tonbo Imaging

Perhaps the most well known of the lot, Tonbo Imaging focusses on improving visibility in the battlefield, a place where dust, smoke, fog, camouflage technology and a number of other factors can hamper it. Based in Bengaluru, Tonbo Imagine designs and manufactures advanced imaging and sensor technology. Their entire product portfolio is too vast to be listed here. However, they incorporate cooled, uncooled and shutterless thermal imaging, micro-optics, high precision inertial stabilisation, real-time video processing, multi-sensor



TONBO
imaging

fusion and more to provide a wide array of imaging solutions to the Indian Armed Forces. These solutions range across driver vision, situational awareness, fire control systems, sights used in weapons, body armour and helmets, recon and targeting, aerial reconnaissance, low visibility navigation and targeting for a wide variety of flying vehicles and much more.

Axio Biosolutions

Based out of Bengaluru, Axio Biosolutions makes products designed